

AZ-900

Learning path 3: Management and governance



Agenda



Learning path 1: Cloud concepts

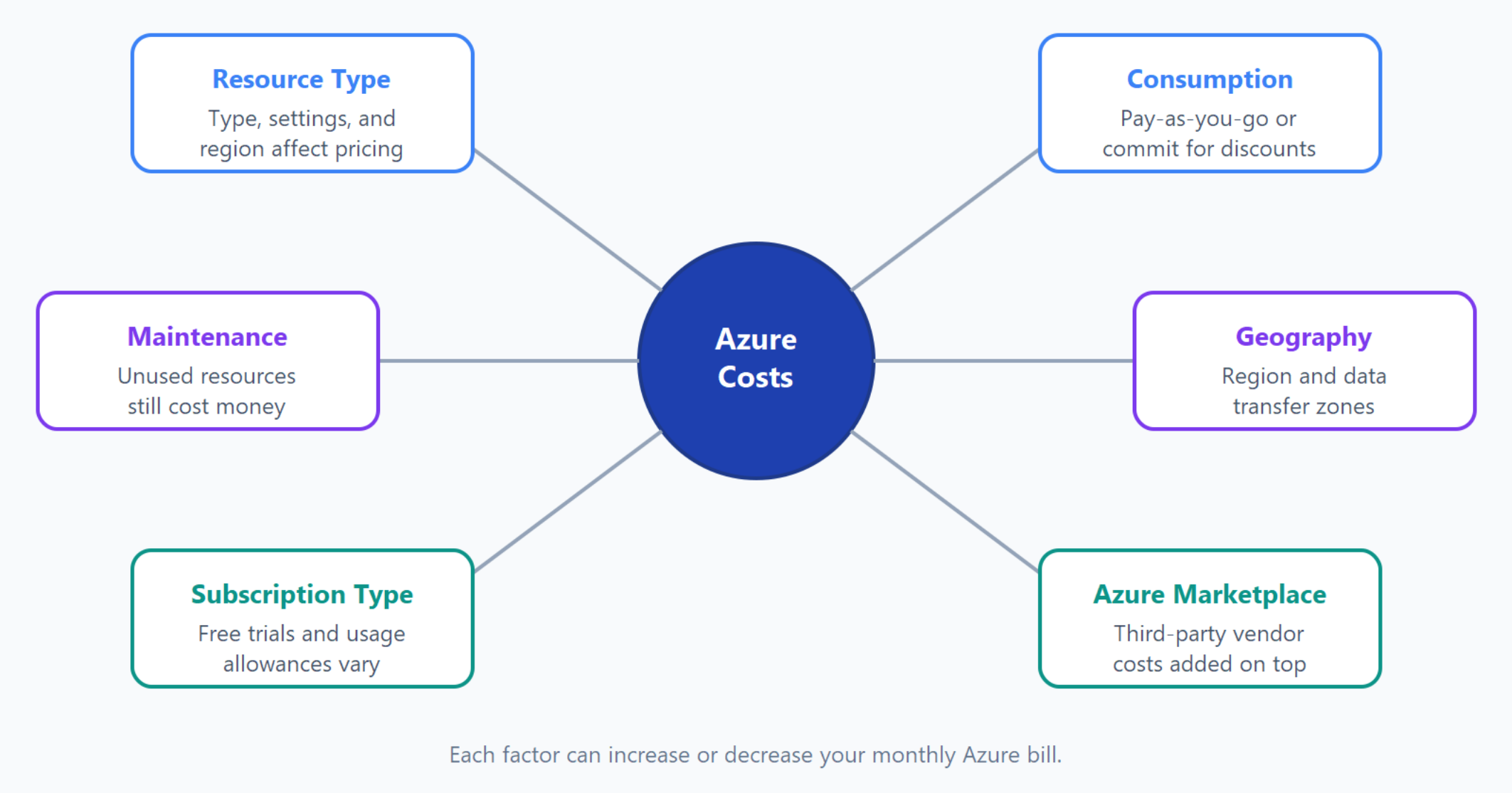
Learning path 2: Azure architecture and services

Learning path 3: Azure management and governance

Describe cost management in Azure



Factors that affect costs in Azure



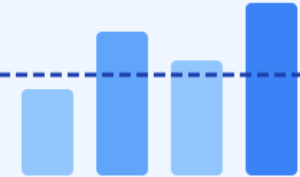
Pricing calculator

- Estimate costs before provisioning resources in Azure
- Build configurations for individual resources or full solutions
- Compare estimates across regions, tiers, and redundancy options
- Use estimates as a baseline for budget approval
- Revisit when architecture changes to keep planned spend aligned
- Information only: nothing is provisioned and no charges are incurred

Microsoft Cost Management

Cost Analysis

View costs by billing cycle, region, resource, or service



Budget line

Cost Alerts

- Budget alerts
- Credit alerts (EA)
- Dept. spending quota

Delivered via email and portal

Budgets

Set spending limits by subscription, resource group, or service type

Configure automation to suspend or modify resources at threshold

Common Tasks

- Spot sudden spend increases
- Find idle resources
- Validate tagging and budgets

Cost Management helps you monitor, allocate, and optimize your Azure spending.

Purpose of resource tags

Resource Management

Locate and act on resources by workload, environment, or owner

Cost Management

Group resources to report costs, allocate cost centers, and budget

Operations Management

Group by availability to formulate service-level agreements (SLAs)

Security

Classify data by security level, such as public or confidential

Governance

Identify resources that align with compliance requirements (ISO 27001)

Workload Automation

Visualize resources in complex deployments and automate tasks

Recommended starter tags: Environment, Owner, CostCenter, Workload

Cost optimization options

- Reservations: commit to specific resources for 1 or 3 years at discounted pricing
- Savings plan for compute: commit to hourly spend amount across eligible compute services
- Spot Virtual Machines: use unused Azure capacity at reduced prices for interruptible workloads
- Decision guide: Reservations for stable workloads, savings plans for flexible compute, Spot for fault-tolerant jobs
- Review choices regularly as workload patterns change over time

Knowledge check



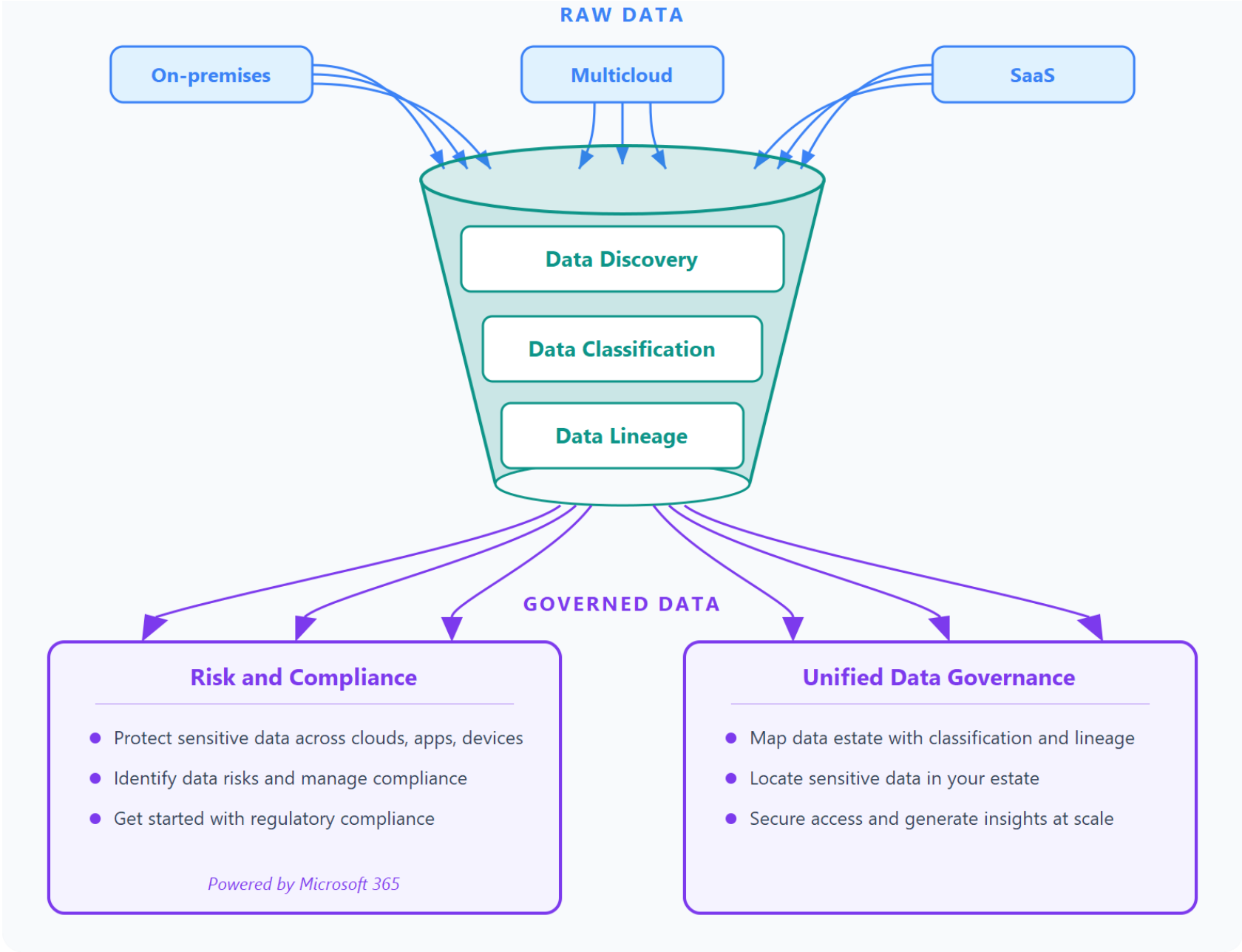
1 Which pricing option is usually best for steady, predictable workloads over a 1- or 3-year term?

- ☐ Spot VMs
- ☐ Reservations
- ☐ Pay-as-you-go only

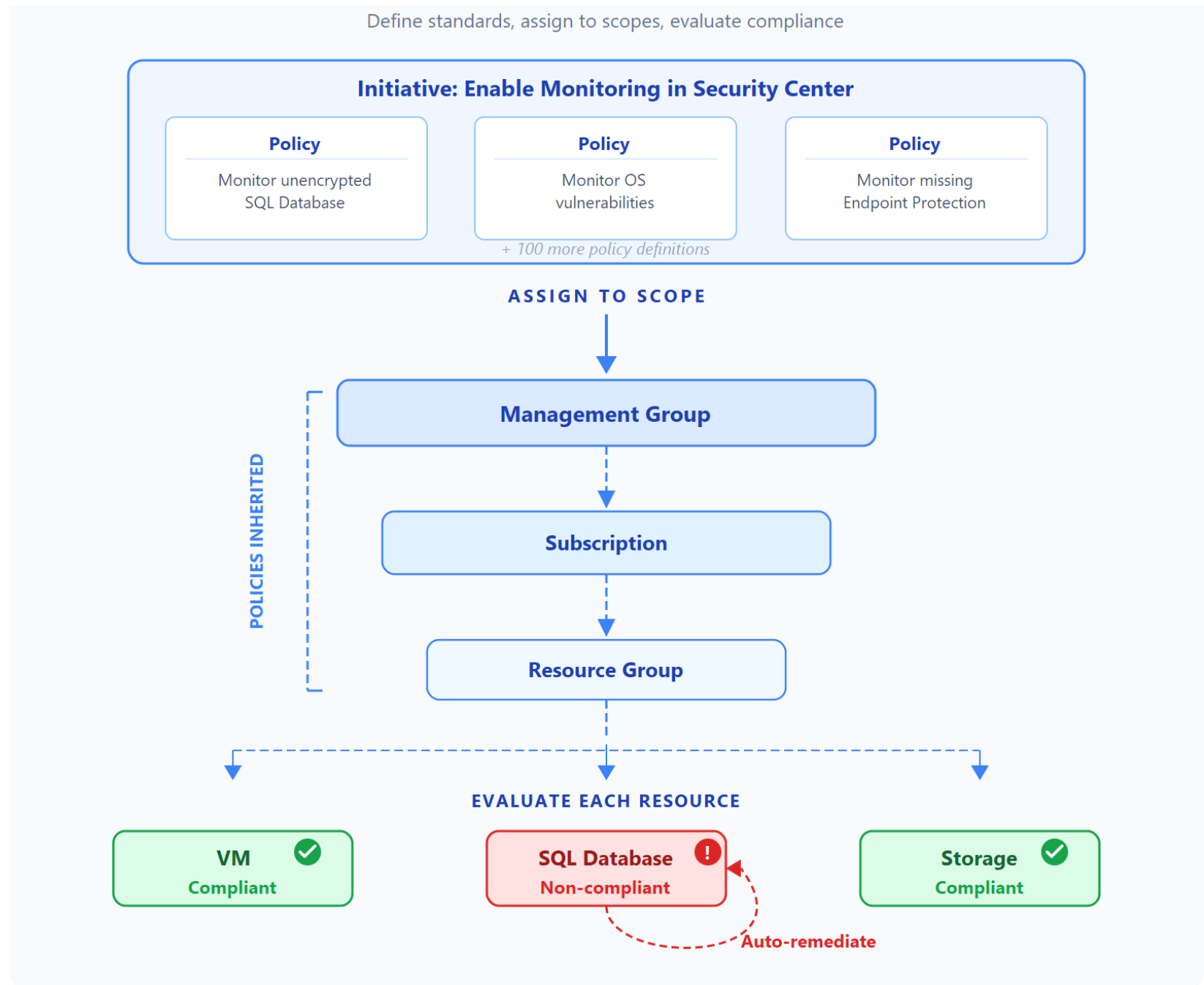
Describe features and tools for governance and compliance



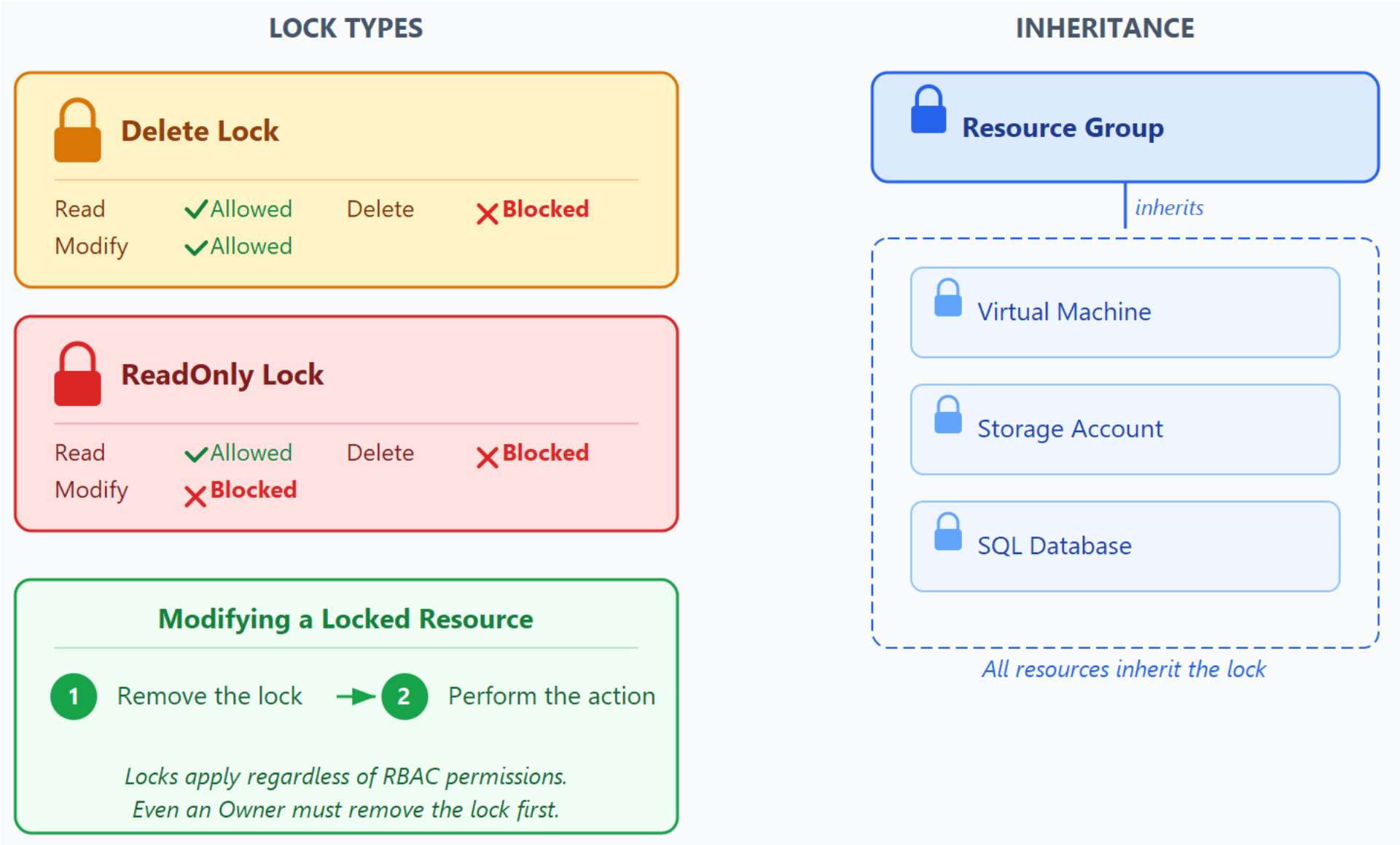
Microsoft Purview



Azure Policy



Resource locks



Service Trust Portal

- Portal for accessing Microsoft security, privacy, and compliance resources
- Contains details about controls and processes that protect cloud services
- Requires authentication with a Microsoft Entra work or school account
- My Library: save and pin documents, set up notifications for updates
- All Documents: single landing place for all available compliance documents
- Reports and documents available for download for at least 12 months

Knowledge check



1 Which service is used to enforce standards and evaluate resource compliance at scale?

- ☐ Azure Policy
- ☐ Azure Advisor
- ☐ Azure Monitor

Describe features and tools for managing and deploying Azure resources



Tools for interacting with Azure

GRAPHICAL Azure portal

Web-based console
to build, manage,
and monitor
cloud resources

ENVIRONMENT Cloud Shell

Browser-based shell
with built-in auth.
No local install
required

COMMAND LINE Azure PowerShell

Cmdlet-based
scripting for
repeatable
automation tasks

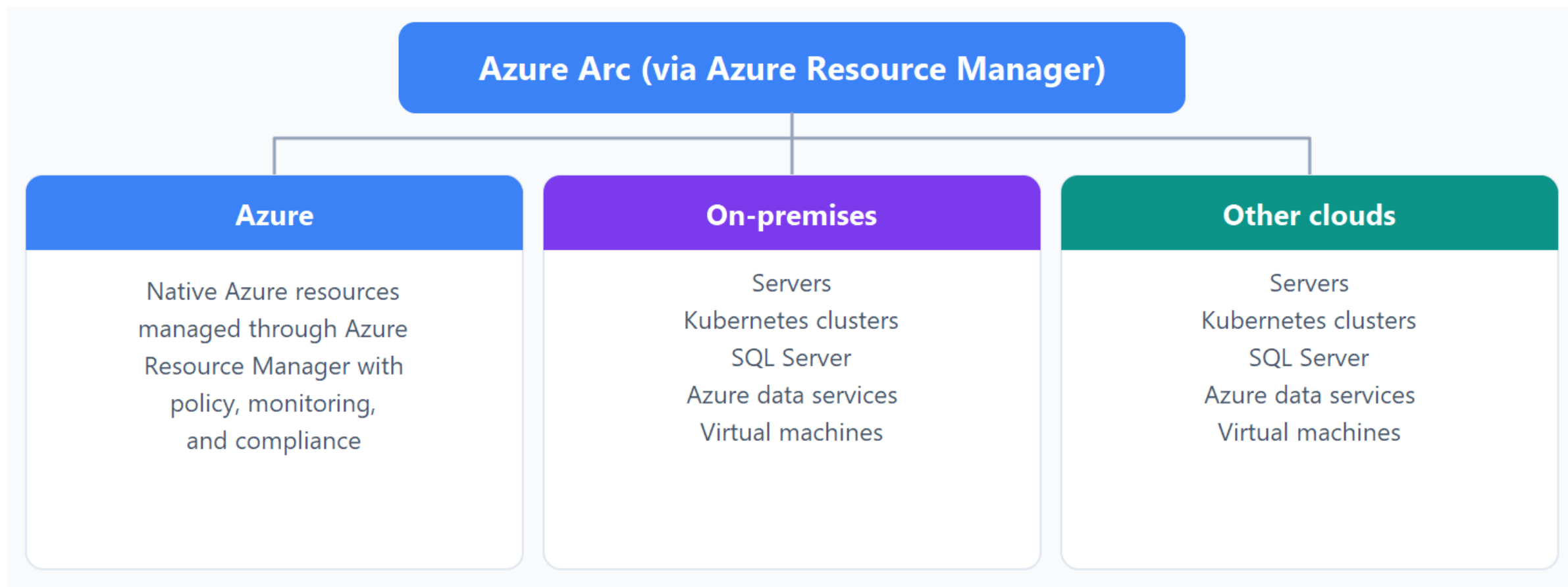
COMMAND LINE Azure CLI

Bash-based
commands with
same capabilities
as PowerShell

AI-ASSISTED Copilot in Azure

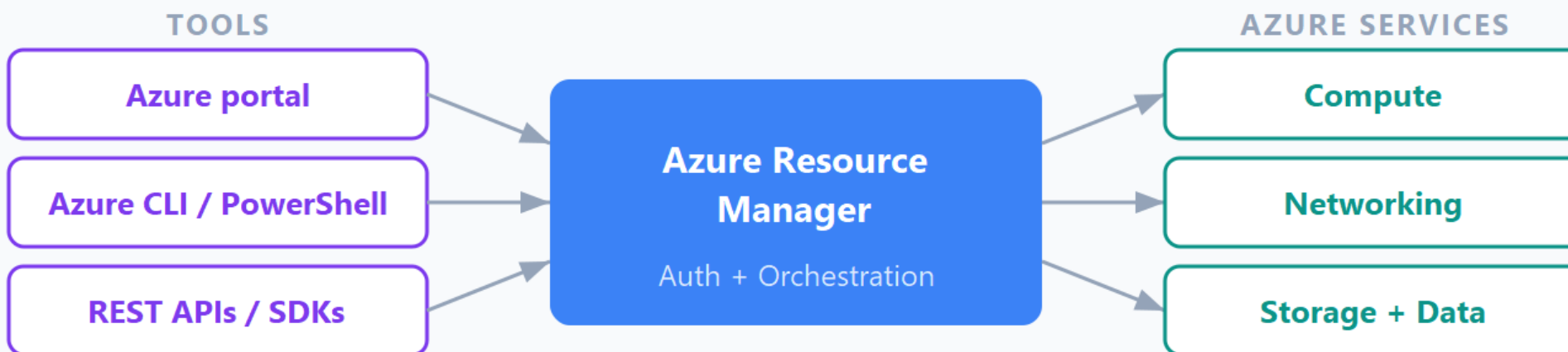
AI assistant for
contextual guidance
in natural
language

Azure Arc – Unified management across environments



Azure Resource Manager, ARM templates, and Bicep

Azure Resource Manager Request Flow



Infrastructure as Code (IaC)

ARM Templates (JSON)

Bicep (Declarative DSL)

Knowledge check



1 Which approach is best aligned to repeatable, version-controlled Azure deployments?

- ☐ Manual portal changes only
- ☐ Infrastructure as Code (for example, ARM/Bicep)
- ☐ Ad-hoc scripts with no source control

Describe monitoring tools in Azure



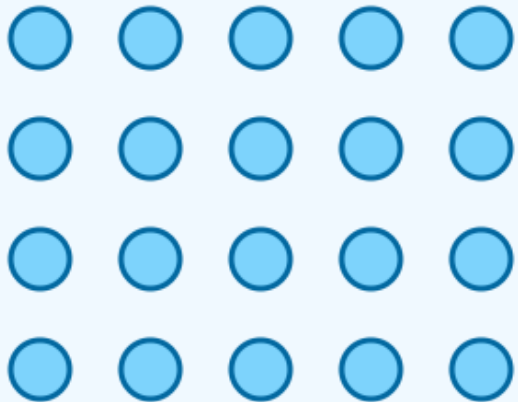
Azure Advisor

- Evaluates Azure resources and makes personalized best-practice recommendations
- Reliability: flags configuration risks to application availability
- Security: detects threats and vulnerabilities that could lead to breaches
- Performance: identifies changes that can speed up applications
- Operational Excellence: suggests workflow and deployment improvements
- Cost: finds ways to reduce Azure spending

Azure Service Health

Azure Status

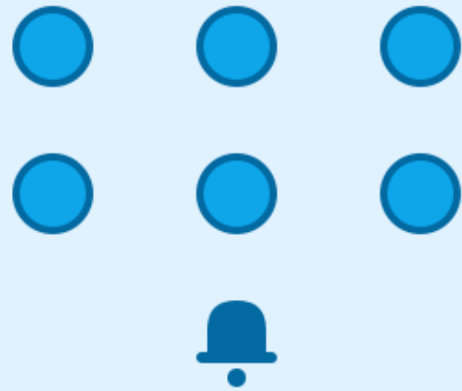
All services, all regions



Global view

Service Health

Your services, your regions



Alerts for your environment

Resource Health

Individual resource



Running normally

Azure Monitor

- Platform for collecting, analyzing, and acting on telemetry across Azure, on-premises, and multicloud
- Log Analytics: write and run queries against collected logs and metrics
- Azure Monitor Alerts: notify when conditions are met with action groups for response
- Metric-based alerts (e.g., CPU above 80%) and log-based alerts (e.g., error patterns)
- Application Insights: monitors web app performance, request rates, response times, and failures
- Synthetic availability tests verify apps are responding during low-traffic periods

Knowledge check

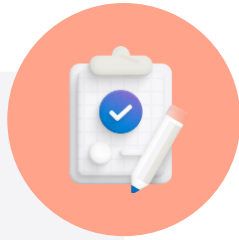


1 Which service should you use to collect metrics/logs and trigger alerts from telemetry?

- ☐ Azure Cost Management
- ☐ Azure Monitor
- ☐ Microsoft Entra ID

Summary

Learning Path 3: Describe Azure Management and Governance



What you learned

- Cost optimization depends on workload predictability: reservations and savings plans for steady usage, spot for interrupt-tolerant workloads.
- Governance at scale relies on policy-driven control and compliance tracking, not manual review.
- Reliable operations require repeatable deployment (infrastructure as code) plus monitoring and alerting with Azure Monitor.

End of presentation



